

Formula Seed for a C++/WebAssembly Physarum Simulator

Project specification seed

May 23, 2026

1 State

The world is a 2D domain $\Omega = [0, W) \times [0, H)$ advanced with time step Δt . Agent i has

$$X_i(t) = (p_i(t), \theta_i(t), E_i(t), m_i(t), \mu_i(t), a_i(t)),$$

where $p_i \in \Omega$ is position, θ_i is heading, E_i is energy, m_i is biomass, $\mu_i \in \{\text{search, exploit, dormant}\}$ is behavior mode, and a_i is alive/dead status.

Food source k has

$$F_k(t) = (f_k, C_k(t), q_k, \sigma_{A,k}, r_k),$$

where C_k is remaining calories, q_k is quality/attractiveness, and r_k is consumption radius.

2 Fields

Trail T_t and food attractant A_t are grid fields. The combined sensing field is

$$S_t(x, y) = w_T T_t(x, y) + w_A A_t(x, y) - w_R R_t(x, y) - w_O O(x, y).$$

Food attractant is

$$A_t(x, y) = \sum_k q_k \frac{C_k(t)}{C_k(t) + C_{1/2} + \varepsilon} \exp\left(-\frac{\|(x, y) - f_k\|^2}{2\sigma_{A,k}^2}\right).$$

Trail deposition by agents is

$$D_t(x, y) = \sum_{i:a_i=1} q_0 m_i(t) \phi_E(E_i) \phi_\mu(\mu_i) K_h((x, y) - p_i(t)).$$

Trail evolves as

$$T_{t+\Delta t} = (1 - \lambda_T \Delta t) (G_{\sigma_T} * (T_t + D_t)).$$

3 Sensing and Motion

Let $u(\theta) = (\cos \theta, \sin \theta)^\top$. Sensor locations are

$$z_i^\delta(t) = p_i(t) + d_s u(\theta_i(t) + \delta \alpha), \quad \delta \in \{-1, 0, +1\}.$$

The corresponding samples are $S_i^\delta = \mathcal{I}[S_t](z_i^\delta)$. The deterministic turn is

$$\Delta\theta_i^{\text{det}} = \begin{cases} +\beta, & S_i^{+1} = \max_\delta S_i^\delta, \\ -\beta, & S_i^{-1} = \max_\delta S_i^\delta, \\ 0, & S_i^0 = \max_\delta S_i^\delta. \end{cases}$$

Search probability can be modeled as

$$P_i^{\text{search}} = \sigma\left(k_E \left[\frac{E_i}{E_{\max}} - \theta_d \right]\right) (1 - \sigma(k_S[S_i^{\max} - \tau_S])),$$

clipped to $[0, 1]$. A weak signal and enough energy make continued search more likely.

Food force is

$$F_i^{\text{food}} = \sum_k \frac{\chi(C_k > 0)c_k}{\|f_k - p_i\|^2 + \varepsilon} \frac{f_k - p_i}{\|f_k - p_i\| + \varepsilon}.$$

Its angle is $\theta_F = \text{atan2}(F_y, F_x)$. Hunger/energy blending uses

$$\rho_i = \text{clip}\left(\|F_i^{\text{food}}\| \left(1 - \frac{E_i}{E_{\max}}\right)^\eta, 0, 1\right),$$

$$\theta_i(t + \Delta t) = \text{wrap}\left((1 - \rho_i)(\theta_i + \Delta\theta_i^{\text{det}} + \xi_i) + \rho_i\theta_F\right).$$

Position update is

$$p_i(t + \Delta t) = p_i(t) + v_0\psi_E(E_i)u(\theta_i(t + \Delta t))\Delta t.$$

4 Energy, Eating, Growth, Death

Energy loss is

$$L_i(t) = \Delta t (c_b m_i + c_m v_i^2 m_i + c_s \mathbf{1}_{\mu_i = \text{search}}).$$

If $d_{ik} = \|p_i - f_k\| < r_k$, food gain is

$$G_{ik}(t) = \min\{C_k(t), \eta_{\text{eat}} q_k \Delta t\}, \quad C_k(t + \Delta t) = C_k(t) - \sum_i G_{ik}(t).$$

Agent energy is

$$\hat{E}_i(t + \Delta t) = \text{clip}\left(E_i(t) + \eta_E \sum_k G_{ik}(t) - L_i(t), 0, E_{\max}\right).$$

Growth occurs only when energy is above a threshold:

$$\Delta m_i = r_g \left[\frac{\hat{E}_i}{E_{\max}} - \theta_g \right]_+ m_i \Delta t, \quad E_i(t + \Delta t) = \hat{E}_i - \kappa_g \Delta m_i.$$

Division can occur when

$$m_i > m_{\text{split}}, \quad E_i > E_{\text{split}}.$$

Death occurs after sustained starvation:

$$E_i(t) = 0 \text{ for } \tau_{\text{dead}} \text{ steps.}$$

5 Adaptive Nutrient Transport Network

Build a graph $G = (V, E)$ from food sources, sampled trail ridges, and optional user-placed nodes. Each edge $e = (u, v)$ has length ℓ_e and conductivity $D_e(t)$. With incidence matrix B ,

$$C_e(t) = \frac{D_e(t)}{\ell_e + \varepsilon}, \quad L_D(t) = B \operatorname{diag}(C_e(t)) B^\top.$$

Pressure solves

$$L_D(t)P(t) = b(t),$$

with a fixed gauge such as one sink pressure set to zero. Edge flow is

$$Q_e(t) = C_e(t)(P_u(t) - P_v(t)).$$

The original Tero-style rule is

$$\dot{D}_e = |Q_e| - D_e.$$

An interactive bounded variant is

$$D_e(t + \Delta t) = \max \left(D_{\min}, D_e(t) + \Delta t \left[\alpha_D \frac{|Q_e|^\gamma}{|Q_e|^\gamma + q_0^\gamma + \varepsilon} - \mu_D D_e(t) \right] \right).$$

The displayed shortest nutrient path can use Dijkstra with

$$c_e^{\text{path}} = \frac{\ell_e}{(D_e + \varepsilon)^\eta} + \lambda_r R_e^{\text{risk}} + \lambda_o O_e.$$

Report total path length

$$L_{\text{path}} = \sum_{e \in P} \ell_e,$$

transport cost

$$J_{\text{transport}} = \sum_{e \in E} \ell_e D_e,$$

and dissipation

$$\mathcal{E}_{\text{diss}} = \sum_{e \in E} \frac{\ell_e}{D_e + \varepsilon} Q_e^2.$$

6 References To Cite In Final Document

Nakagaki et al. (2000), Tero et al. (2007), Jones (2010), Bonifaci et al. (2012/2013), Kay et al. (2022), and MDN/Emscripten WebAssembly documentation.